

MVP OVERVIEW / AUGUST 2026

CA

# ChargeAssert

Financial release testing  
for EV charging.

Catch billing defects before they reach customers or  
revenue systems.

**SELIM ABOULEILA**

Independent public portfolio project  
Public and synthetic data only

## 01 / MVP OVERVIEW

# The product

ChargeAssert is a test harness for EV-charging software. Before a new version goes live, it replays realistic charging sessions and checks that every session produces the correct financial result.



**A release can look technically healthy while this financial chain is wrong.**

## What it catches

- + Missing charging records
- + Duplicate customer charges
- + Incorrect energy or duration
- + Wrong tariff or tariff version
- + Retry and idempotency errors
- + Late or out-of-order events

### PROTECT CUSTOMERS

Stop duplicate and inflated charges before deployment.

### PROTECT REVENUE

Expose missing or understated billable records.

### MOVE FASTER

Give engineering a reproducible cause, not just an error count.

### THE BUSINESS VALUE

Shift financial defect discovery from post-production reconciliation to a clear release decision before deployment.

Primary users: QA and platform engineering, revenue assurance, OCPP / OCPI integration teams, and charging-platform vendors.

## 02 / MVP OVERVIEW

# How it works

ChargeAssert sends the same deterministic scenario manifest through a trusted baseline and the candidate release, then compares both outputs with an independent expected result.

**ONE MANIFEST. TWO RELEASES. ONE FINANCIAL VERDICT.**

The same seed, event clock and inputs make every failure reproducible.

**BASELINE VS CANDIDATE**

Both releases receive identical events, IDs and timestamps. The comparison exposes what changed in the candidate.

**INDEPENDENT ORACLE**

Expected energy, tariff and cost are calculated separately. An old baseline defect cannot silently become acceptable.

## Black-box by design

The MVP tests controlled mock systems. A future customer can replace the mock adapter with approved staging endpoints. ChargeAssert needs inputs, outputs and tariff rules - not access to proprietary source code.

**WHY NO MACHINE LEARNING?**

The MVP uses deterministic rules because financial checks need known expected outcomes, clear explanations and exact reproduction. ML can be explored later with labelled operational data.

## 03 / MVP OVERVIEW

# A focused MVP

The first release proves one complete financial quality gate. Everything else stays outside the boundary until the core workflow is reliable and useful.

## INCLUDED

- + Synthetic OCPP and OCPI-shaped events
- + Baseline and candidate comparison
- + Six flagship failure scenarios
- + Independent financial assertions
- + Modeled EUR exposure
- + GitHub PASS / FAIL report
- + Reproducible Databricks pipeline

## NOT IN THE MVP

- + Real card or payment processing
- + Bank, PSP, ERP or settlement integration
- + Full OCPP / OCPI certification
- + Production monitoring or recovery
- + Every tariff, tax system or currency
- + Machine learning
- + Confidential operator data

## MVP outputs

### VERDICT

GitHub PASS /  
FAIL

### CAUSE

First divergence

### IMPACT

Overbilling vs  
leakage

### EVIDENCE

Reproducible run  
manifest

### DEFINITION OF DONE

A deliberately faulty candidate fails, the report explains why, and the corrected candidate passes the exact same scenario manifest.

## 04 / MVP OVERVIEW

# Six failures that matter

The scenario library stays intentionally small. Each case represents a defect with a clear customer or revenue consequence and a deterministic expected outcome.

**01 MISSING CDR**

A completed session never becomes billable.

**02 DUPLICATE CHARGE**

A retry creates a second financial record.

**03 WRONG ENERGY**

Meter order, reset or units distort kWh.

**04 WRONG TARIFF**

The incorrect effective price is applied.

**05 RETRY FAILURE**

An HTTP error breaks idempotency.

**06 LATE EVENT**

Out-of-order data changes the final state.

## Core financial checks

- + One completed billable session produces one final CDR.
- + CDR energy matches validated meter evidence.
- + The correct effective tariff is applied.
- + The final amount matches the independent calculation.

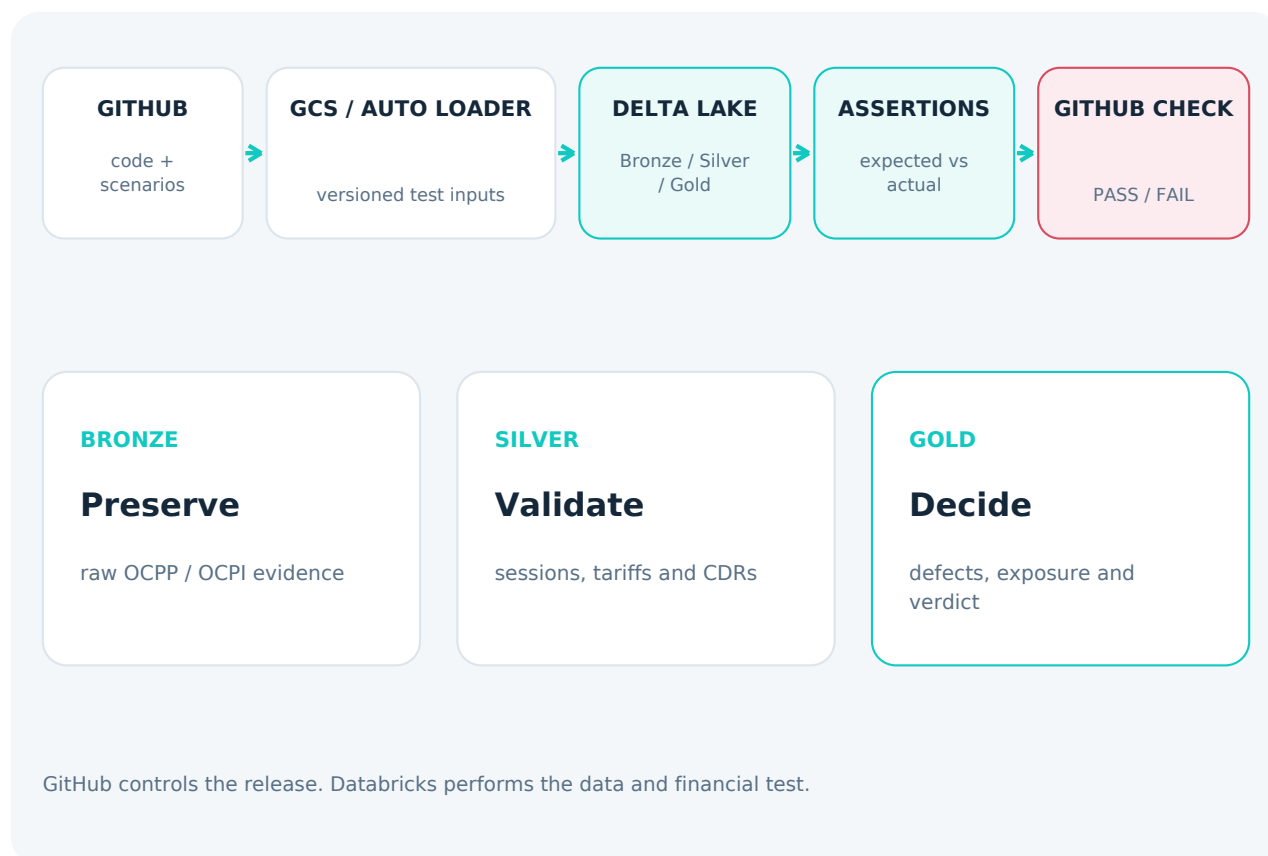
**ONE  
IMPORTANT  
DISTINCTION**

The same event ID delivered twice is a transport duplicate to deduplicate. Two different CDR IDs for one session are a business defect to preserve and report.

## 05 / MVP OVERVIEW

# A simple Databricks architecture

GitHub is the control plane. Databricks is the execution plane. Public-data snapshots and synthetic events move through a compact lakehouse flow and return one release verdict.



## INTERFACES

- + OCPP subset for charger-to-platform events
- + OCPI-aligned Sessions, Tariffs and CDRs
- + GitHub Actions and Declarative Automation Bundles

## PUBLIC INPUTS

- + ACN-Data for realistic charging behavior
- + French IRVE data for static station profiles
- + Synthetic tariffs and controlled fault injection

### DATA PROVENANCE

ACN behavior and French IRVE station metadata are combined synthetically. They are not presented as real French charging transactions or actual operator tariffs.

## 06 / MVP OVERVIEW

# The release gate

The product moment is a pull request that fails for a financial reason, shows the first divergence and turns green after the exact defect is fixed.

chargeassert / release-gate

## RELEASE GATE: FAILED

|                          |            |
|--------------------------|------------|
| Sessions tested          | 50,000     |
| Missing CDR delta        | +0.42%     |
| Duplicate charges        | +0.18%     |
| Modeled monthly exposure | EUR 24,380 |

### FIRST DIVERGENCE

An OCPI retry after HTTP 500 created a second CDR for the same logical charging session.

Baseline: 8f6c2a1  
Candidate: b319ad4  
Scenario: retry\_after\_500

Illustrative result. Counts and financial figures are modeled, not actual losses.

## What the report must answer

### WHAT BROKE?

The first event, retry or tariff decision where behavior diverged.

### WHO IS EXPOSED?

Customer overbilling and operator underbilling shown separately.

### CAN WE REPLAY IT?

Run ID, seed, SHAs, snapshot hashes and a reproduction command.

### MODELED EXPOSURE

defect-rate delta x assumed monthly sessions x assumed impact per affected session

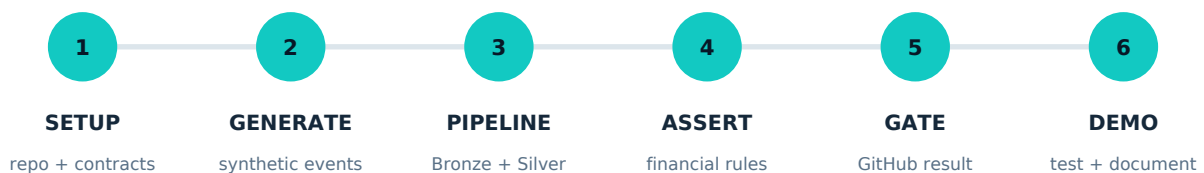
All projections show their assumptions. They are modeled exposure - never actual loss, recovered revenue or proven savings.

## 07 / MVP OVERVIEW

# Build the MVP in six weeks

The first six weeks deliver a complete portfolio demonstration. The next phase should be driven by a real staging partner, not by adding complexity for its own sake.

## SIX-WEEK MVP



## NEXT PHASE

Connect one consenting staging system, import real test tariff contracts, extend the evidence pack and measure false positives and investigation time.

## HONEST BOUNDARIES

- + Not an OCPP / OCPI certification authority
- + Not a card-payment or settlement platform
- + Not proof of real financial savings

## SOURCE SET

[Open Charge Alliance - OCPP](#)  
[Official OCPI repository](#)  
[Caltech ACN-Data](#)  
[French national IRVE dataset](#)  
[Databricks Auto Loader](#)  
[Databricks medallion architecture](#)

**THE MVP  
PROMISE**

**Catch missing, duplicate and incorrectly priced EV-charging records before the associated release reaches production.**

Independent educational and portfolio project. No confidential employer, operator or customer data. All monetary outcomes are synthetic and modeled.